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Laser transmission welding of Poly(vinyl chloride) and polyamide 66 based on green femtosecond laser surface modification

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Highlights

- GFL modification induced oxygen-containing functional group on the PVC surface.
- The modified layer improved miscibility and welding performance of PVC/PA66.
- Response surface methodology was used to optimize the process parameters.
- The experimentally determined optimal joint shear strength was 8.9MPa.

Outline

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Funding

Abstract

To improve the bonding strength of unweldable poly(vinyl chloride) (PVC) and polyamide 66 (PA66), a surface treatment method using green femtosecond laser (GFL) to modify PVC before welding was proposed in this study. The experiments of GFL surface modification, 1940nm laser transmission welding (LTW), mechanical properties test and related characterization were carried out. The effects of GFL parameters on the physical and chemical properties of the modified surface and the bonding strength of the dissimilar joints were investigated in detail. The bonding mechanism of PVC/PA66 joints was analyzed comprehensively. The results showed that laser modification induced the deposition of many oxygen-containing functional groups on the PVC surface. These oxygen-containing functional groups form a highly polar modified layer on the surface, thereby improving the miscibility and welding performance of PVC/PA66. Finally, the modification and welding parameters were optimized based on the response surface methodology (RSM), and the maximum shear strength was 8.9MPa.

Keywords

Bonding strength; Laser surface treatment; Laser transmission welding; Miscibility

CRediT authorship contribution statement

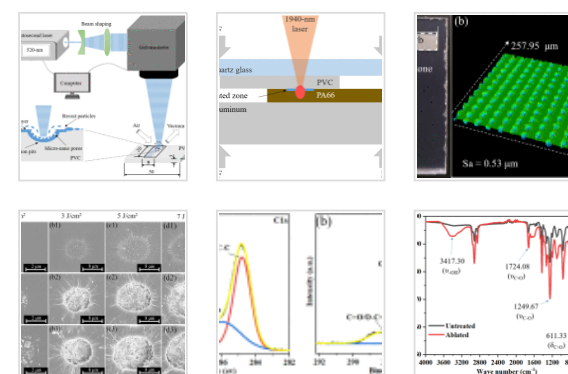
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1. Introduction

Polymers have excellent physicochemical properties and are widely used in the automotive, electronics, aerospace, and medical fields. In practical industrial applications, functional products can only be manufactured through the optimal integration of dissimilar materials and use of appropriate bonding techniques. Laser transmission welding (LTW), with its high efficiency and non-contact advantages, has been widely used in the field of polymer connection in recent years [1], [2], [3]. Because of the low polarity and strong surface inertness of common polymers, they cannot be easily bonded with other materials. In order to improve the bonding strength of polymers with dissimilar materials, researchers have tried various

techniques to modify the surface of polymers. In 2016, Liu et al. used maleic anhydride (MAH) and glycidyl methacrylate (GMA) to graft modify polyethylene (PE), which improved the welding performance between unweldable PE and PA66, and increased the bonding strength to 10.7 and 5.3MPa, respectively [4]. In 2017, Liu et al. used magnetron sputtering technology to sputter a 20- μm -thick aluminum film on the surface of PA66 as a transition layer, increasing the bonding strength between PA66 and PVC from 0MPa to 4.0MPa [5]. In 2017, Yu et al. used low-temperature plasma to desurface TPE, which improved the welding performance of TPE and PMMA (from 4.5MPa to 16.5MPa) [6]. However, these methods have some obvious shortcomings, such as the introduction of new materials (coating), complex processes (chemical grafting), difficulty in accurately controlling the action area (plasma modification) and low efficiency. These drawbacks limit their application in the field of joining dissimilar polymers and their composites.

Pulsed laser direct ablation (PLDA) is a high-efficiency, environmentally friendly, and precisely controllable modification technique. A focused high-energy laser beam is employed to irradiate the surface of a polymer, leading to photochemical and photothermal reactions that alter the material's surface morphology and chemical composition [7], [8], [9], [10], [11]. This technique has been used for polymer surface modification and significant improvements in the adhesive strengths between modified dissimilar materials, e.g., polymer/polymer and polymer/metal, have been reported. Akman and Li et al. used ultraviolet nanosecond and infrared picosecond pulsed lasers to ablate carbon-fiber-reinforced polymers (CFRP). Polar functional groups, e.g., C=O, O=C-O, and -OH, were generated on the surface. The adhesive molecules formed strong chemical bonds with the functional groups (e.g., -COOH), resulting in enhanced CFRP/CFRP [12] and CFRP/Al [13] bonding strengths. Du et al. used ultraviolet nanosecond and infrared picosecond pulsed lasers to modify polyimide (PI). The changes in the physical morphology and chemical properties increased the surface energy of PI, thereby enhancing its bonding strength with copper after gluing [14]. In addition, PLDA was first applied in LTW by Liu et al. They used ultraviolet femtosecond pulse laser to ablate polycarbonate (PC), which induced the deposition of abundant polar oxygen-containing functional groups on the surface, thereby enhancing the miscibility and welding strength between PC and 30% glass-fiber-filled polyamide 66 (PA66GF30) [15].

In PLDA of polymers, factors such as the laser wavelength, energy density, pulse number, material characteristics and environment significantly affect the results [10], [16], [17], [18], [19], [20]. For example, the modification of PI with 1064nm laser could obtain higher content of oxygen-containing functional groups on surface than 355nm laser [10]. The application of PLDA in LTW has been reported in the literature [15], but only the welding effect under fixed modified laser parameters has been investigated. To study the effect of different modified laser parameters on the welding results, the influence of GFL with different energy density and pulse number on LTW was studied, and the feasibility of GFL modification to improve the miscibility and welding performance of unweldable PVC and PA66 was verified by relevant experiments. In addition, the modification and welding parameters were optimized by response surface methodology (RSM) to obtain the optimal bonding strength.

2. Experimental designs

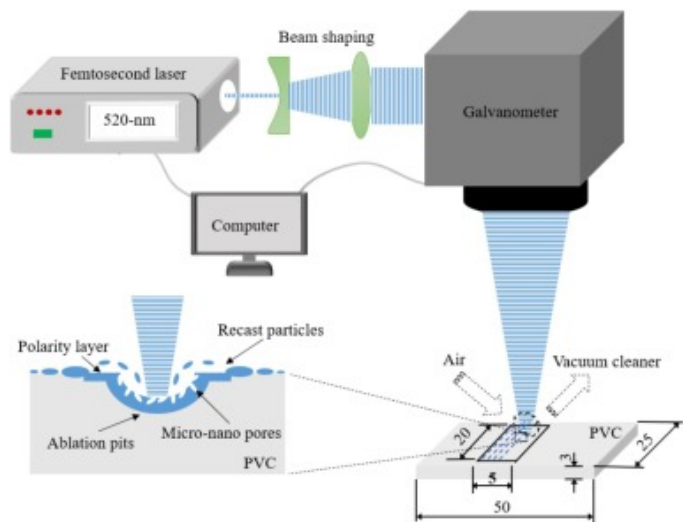
2.1. Materials and surface modification

The PA66 and PVC sheets used were of dimensions $50 \times 25 \times 2 \text{ mm}^3$ and $50 \times 25 \times 3 \text{ mm}^3$, respectively, and were produced by injection molding. The physical properties of the materials are shown in Table 1, where the transmittance is detected by an ultraviolet–visible infrared spectrophotometer (Model: Lambda 1050+). To remove surface impurities and grease, the sheets were wiped with 99.99% alcohol. After evaporation, they were placed in a drying oven 12h before the femtosecond laser modification experiments.

Table 1. Physical properties of PVC and PA66.

Materials	Density (g/cm ³)	Tensile strength (MPa)	Transmittance	
			520nm	1940nm
PVC	1.4	60	85.0%	71.3%
PA66	1.14	72.5	42.6%	3.4%

A GFL pulsed laser with a wavelength of 520nm was used for the PVC surface modification experiments. The maximum average power of the laser was 7W. The laser repetition rate was 250kHz, and the pulse width was 300fs. The focal diameter of the laser beam was approximately 20 μ m, and the maximum scanning speed was 12m/s. During the modification process, the defocus level was set at 0mm (the focal point was on the PVC surface). The scanning route was 200 straight lines (line length of 20mm) with an interval of 25 μ m within the modified range (20 $\times$ 5 mm²), as shown in Fig. 1. The effect of PVC fragments on the modification process was eliminated by blowing compressed air at 0.02MPa from one side to the modification area. A vacuum cleaner was used on the other side to remove the exhaust air. Fig. 1 shows a schematic diagram of the femtosecond laser modification system. After the laser modification experiments, the PVC samples were blown with 0.25MPa air to prevent the ablated residues from affecting the surface chemistry.



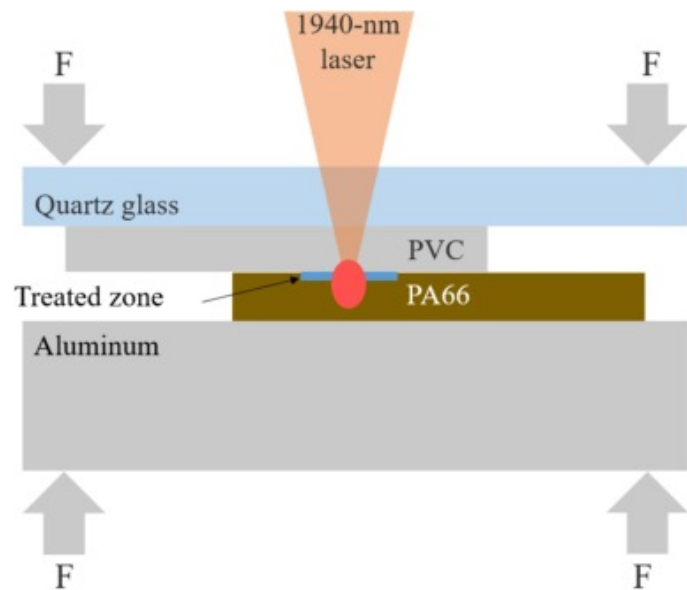
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Fig. 1. Schematic diagram of green femtosecond laser direct ablation equipment. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2.2. Laser welding

For lap welding, PVC and PA66 were set in a fixture in which quartz glass with a high transmittance to a 1940nm wavelength laser was pressed on the PVC; the PA66 was placed on an aluminum plate, as shown in Fig. 2. The system was equipped with a pair of 25mm diameter air cylinders with guide rods to move the glass and provide the pressure (set to 0.4MPa) required during welding. The light source used for welding was a 1940nm continuous thulium-doped fiber laser, with a maximum output power of 20W and a minimum spot diameter of 500 μ m. All samples were welded by single-line contour welding. Due to the high transmittance of the PVC sample to the 1940nm laser (>60%), most of laser passed through the PVC, and part of it was absorbed by the bottom PA66 and converted into heat. Some of this deposited heat was then conducted to the PVC, and effective material welding was achieved, similarly to the case for typical laser transmission welding [21], [22]. It is worth noting that the PVC mentioned here is a material modified by femtosecond laser, and unmodified PVC/PA66 cannot be welded directly.



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Fig. 2. Schematic diagram of laser transmission welding.

2.3. Mechanical performance testing and characterization

After welding, the hybrid joints were mounted on a shear fixture, and a universal testing machine was used to test its shear strength. During the test, the loading speed was set at 2mm/min. After fracture of the welded samples, the maximum failure shear tensile force of each sample was recorded, and the shear strength of the sample was calculated according to **Eq. (1)**.

$$\sigma = F/(L \times W) \quad (1)$$

where σ is the shear tensile strength (MPa), F is the shear tensile force (N), L is the welding length (20mm, fixed value), and W is the welding width (mm). The effect of laser modification and the failure mechanism of hybrid joints were investigated by using laser scanning confocal microscopy (LSCM) to examine the overall microstructure and roughness of the modified surface. Scanning electron microscopy (SEM) was used to observe the local microstructure of the laser-modified region and the fracture surface of the weld joints. X-ray photoelectron spectroscopy (XPS) and Fourier transform infrared spectroscopy (FTIR) was used to determine the chemical compositions of the material surface and the fracture surface of the hybrid joints. The water contact angle (WCA, droplet volume: 2 μ L) before and after ablation was measured using a water droplet angle tester to investigate the wettability on surface.

2.4. RSM

In this study, Design-Expert 11 software was used for RSM experimental design and analysis. According to the experimental design principles of central composite design, a four-factor and five-level experimental scheme was developed. The RSM factors and levels are given in [Table 2](#).

Table 2. RSM factors and levels.

Factors	Levels				
A-GFL Energy Density (J/cm ²)	1	2.5	4	5.5	7
B-GFL Pulse Number	2	3	4	5	6
C-Welding Power (W)	6	9	12	15	18

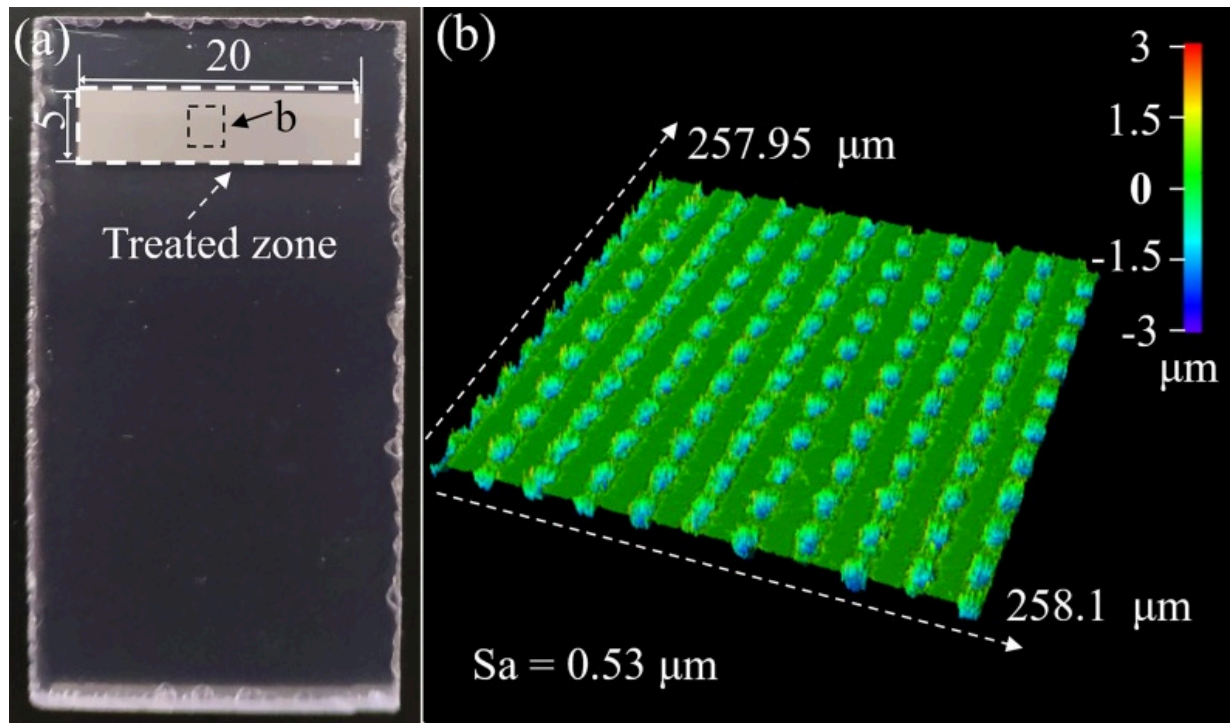
Factors	Levels				
A-GFL Energy Density (J/cm ²)	1	2.5	4	5.5	7
D-Welding Speed (mm/s)	1	1.4	1.8	2.2	2.6

3. Results and discussion

3.1. Laser ablation modification results and analysis

3.1.1. Surface morphology

[Fig. 3\(a\)](#) and [\(b\)](#) show a photograph and a local LSCM image of the modified PVC, respectively. The boundaries of the rectangular modified region are clear in [Fig. 3 \(a\)](#), and the dimensions are consistent with the set value (5×20mm²). In the magnified image in [Fig. 3 \(b\)](#), a regular array of pits can be observed, with a surface roughness value (Sa) of 0.53μm. This indicates that femtosecond laser processing gives precise control over the modified region, which is difficult to achieve with other modification techniques such as plasma treatment or chemical grafting.



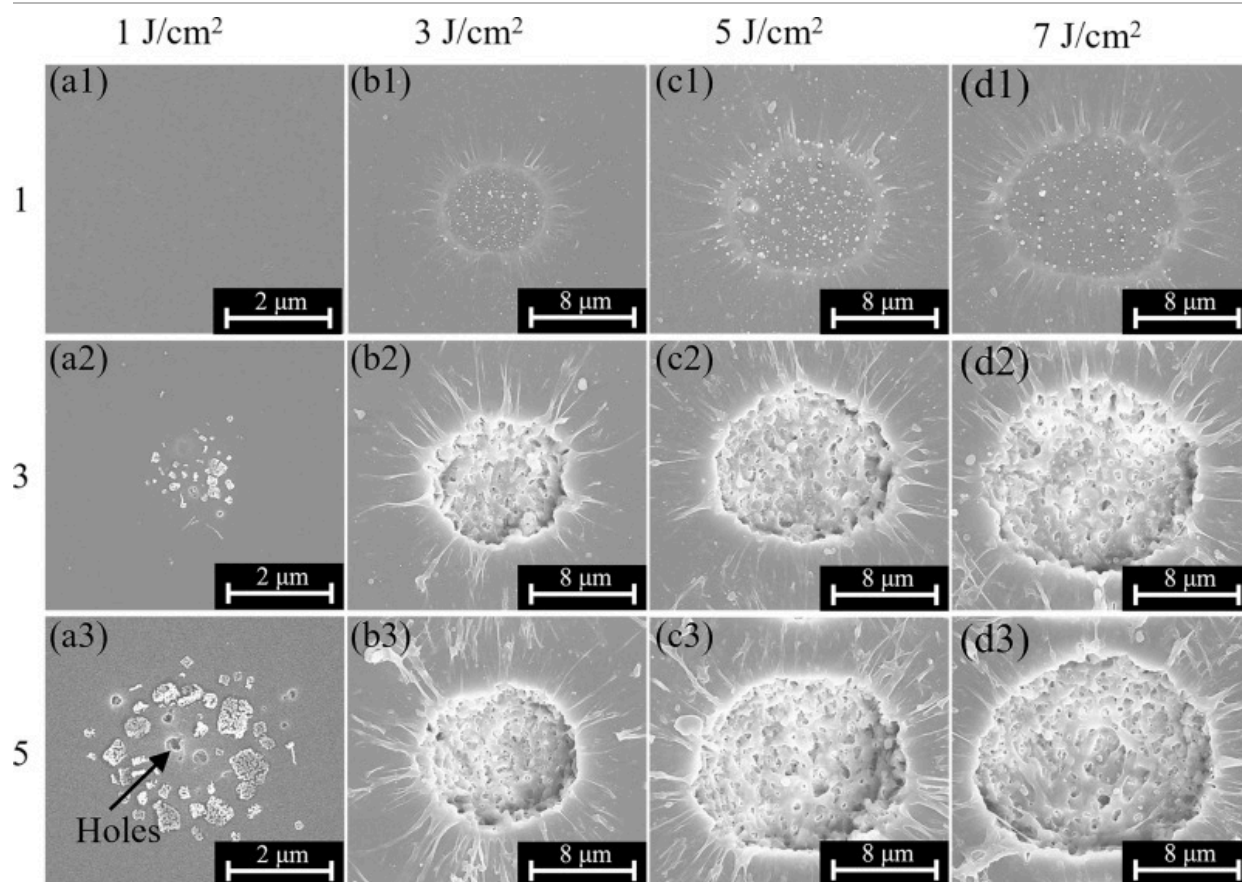
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Fig. 3. PVC after modification: (a) photograph; (b) local LSCM image.

Laser ablation modification experiments using lasers with different energy densities and pulse numbers were conducted on PVC to investigate the effects of the laser parameters on the microstructure of the PVC surface and to enable selection of a reasonable range of ablation parameters. The microstructure after ablation is shown in Fig. 4. Under the conditions of single-pulse ($B=1$) and a low energy density ($A=1 \text{ J/cm}^2$), no change was observed on the PVC surface (Fig. 4(a1)). When $A \geq 3 \text{ J/cm}^2$ and $B=1$, a blurry disk structure appeared in the ablation region. This is a trace left by material melting and it expands with increasing energy density (Fig. 4(b1)–(d1)). However, no obvious modification traces resulting from decomposition reactions are found in the ablation region, which indicates that low-pulse laser ablation ($A=1 \text{ J/cm}^2$) was not effective for PVC modification. Multiple-pulse laser ablation could lower the threshold of material photodecomposition reactions, which enables ablation modification with low-energy–

density lasers [23]. However, as shown in Fig. 4(a2) and (a3), only a small number of voids and splatters appeared on the material surface, which indicates weak photodecomposition reactions; this also suggests that low-energy-density ($A=1 \text{ J/cm}^2$) and multiple-pulse-number ($B \geq 3$) lasers were not effective in ablative modification of PVC. When A and B were increased to 3 J/cm^2 and 3, respectively, the material surface showed circular pits with a diameter approximately $8 \mu\text{m}$ filled with microcavities and radial stripes at the edges. These pits are typical microstructural features left by the melting, decomposition, and ejection of polymer materials, and indicate effective ablative modification of the material. The pit size showed no significant change with increasing pulse number but rapidly increased at higher energy densities. For example, at $A=3 \text{ J/cm}^2$ and $B=3$, and $A=7 \text{ J/cm}^2$ and $B=3$, the pit diameters increased to ~ 9 and $\sim 16 \mu\text{m}$, respectively. In summary, appropriate laser parameters are pivotal for achieving effective ablation. In this study, effective modification of PVC by using GFL ablation was achieved when the energy density was $\geq 1 \text{ J/cm}^2$ and the pulse number was ≥ 3 .



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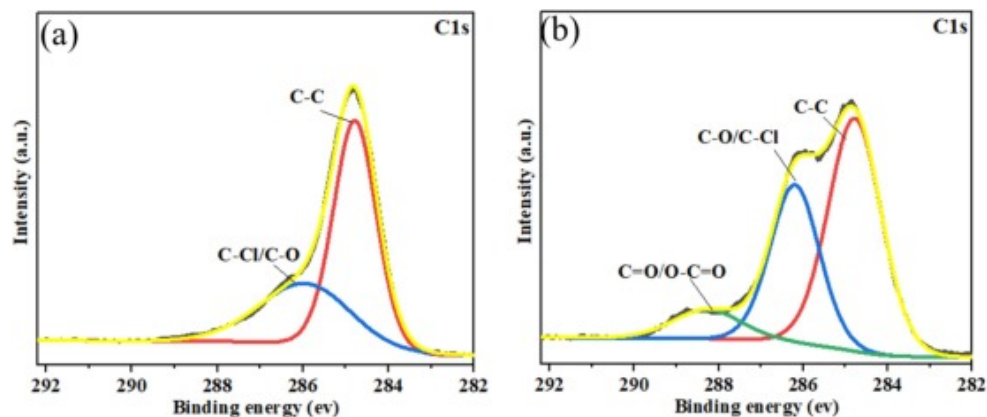
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Fig. 4. SEM images of modified PVC under different energy densities (1, 3, 5, 7 J/cm²) and pulse numbers (1, 3, 5).

3.1.2. Surface chemical analysis

The XPS spectra of the pristine and modified PVC are shown in Fig. 5. Unlike the pristine PVC, the XPS spectra of the modified PVC show peaks corresponding to C=O/O-C=O bonds in the C1s spectrum at 288.8 eV (Fig. 5(a), (b)) [24], [25]. In addition, compared with the previous study of 355 nm laser ablation of PC [15], FTIR shows that the modified sample has an obvious new absorption peak between the wave number of 3300–3500 cm⁻¹ (Fig. 6), which is caused by the

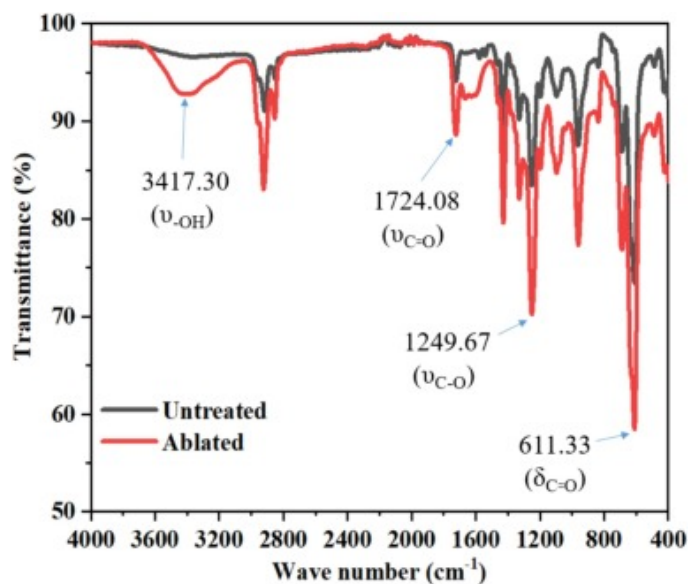
stretching vibration of –OH, indicating that the content of –OH groups on the surface of the modified PVC sample increases sharply [26], [27], [28]. The XPS and FTIR results indicate that GFL ablation of the PVC surface generates many oxygen-containing functional groups. This is probably the result of the synergistic effects of photothermal [8], [10], [29] and photochemical reactions [29], [30], [31], [32], [33], [34]. Fig. 7 shows the bond energy of PVC chemical bonds and the photon energy corresponding to different wavelength photons. Obviously, the single photon energy of the used laser for modification is not enough to break the C–C/C–H/C–Cl chemical bond in PVC molecules. Therefore, during the modification process, PVC absorbed multiple 520nm photon energy through nonlinear absorption (mainly two-photon absorption), and then broke its carbon bonds. This process produced free radicals on the material surface, which then reacted rapidly with oxygen atoms in the air to form –OH/C=O/O–C=O bonds. The possible related photochemical reaction mechanism is shown in Fig. 8. Due to the deposition of a large number of oxygen-containing functional groups on the material surface after laser ablation, the polar force ($\delta p = 7.8 \text{ MPa}^{-1/2}$) of the modified PVC would be improved. According to the Hansen solubility similarity-miscibility theory, laser ablation reduces the difference ($\Delta \delta p$) between the polar forces of PVC and PA66 ($\delta p = 9.8 \text{ MPa}^{-1/2}$), thereby reducing their solubility vector distance (Fig. 9), which means that the mixing enthalpy of the two molten polymers decreases and their miscibility improves [35], [36]. This rule is called “like seeking like,” which reflects if the type and size of the components of intermolecular forces are similar, the mutual solubility is excellent. In essence, the reason for the improvement of the bonding strength of the hybrid joints after laser modification is the enhancement of the diffusion and entanglement between the dissimilar molecules, that is, the increase of van der Waals attraction, resulting in the enhancement of their fusion and bonding ability. In conclusion, the improvement of miscibility caused by the decrease of $\Delta \delta p$ could enhance the weldability of PVC/PA66 to a certain extent. Interestingly, due to the reduction of oxygen atoms involved in photochemical reactions, the content of oxygen-containing functional groups on the surface of PI induced by laser ablation in air was significantly higher than that in argon [10]. It can be speculated that laser ablation in air (or oxygen) can make the dissimilar polymers obtain better welding performance, but more related experiments need to be carried out to verify in the future.



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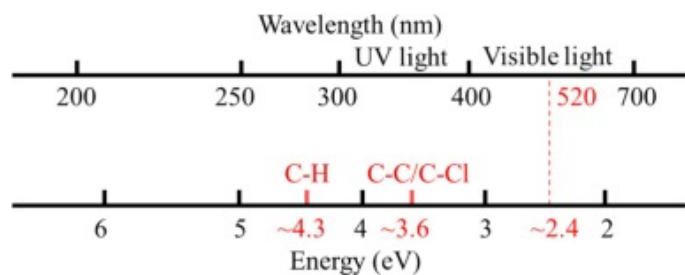
Fig. 5. XPS spectra of PVC: (a) pristine; (b) after laser ablation ($A=4 \text{ J/cm}^2$, $B=4$).



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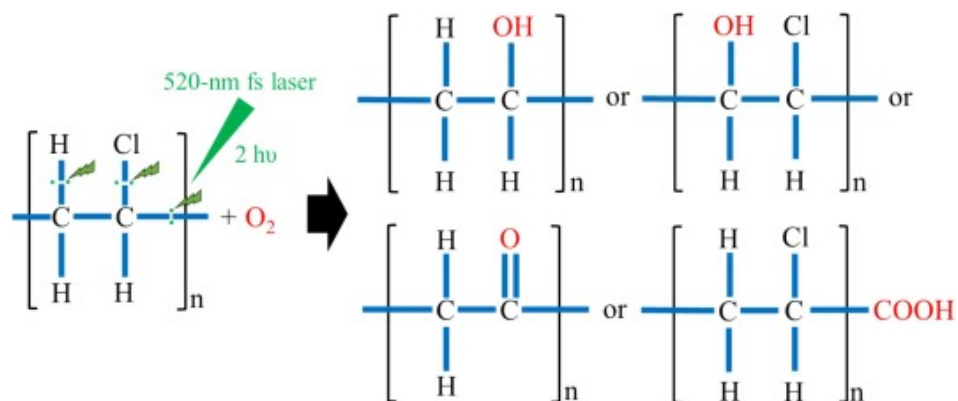
Fig. 6. FTIR spectrum of the pristine and ablated PVC.



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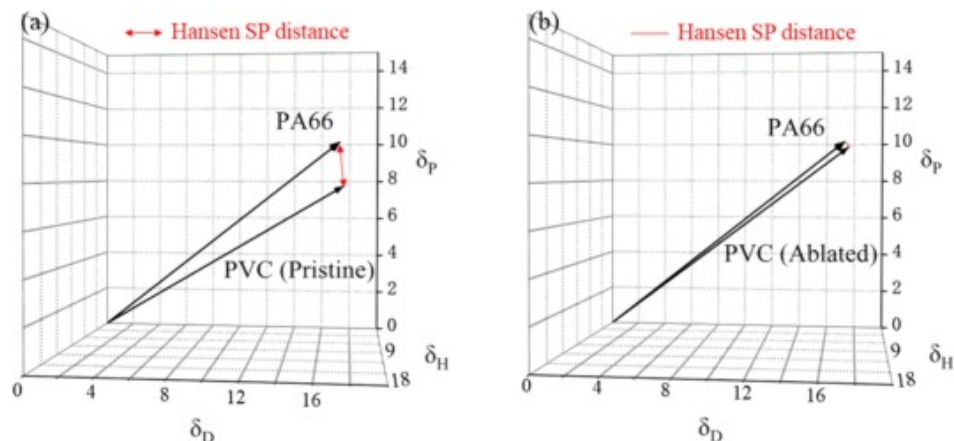
Fig. 7. Energy diagrams corresponding to different wavelengths of photons and different chemical bonds.



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Fig. 8. Mechanism diagram of femtosecond laser ablation of PVC.



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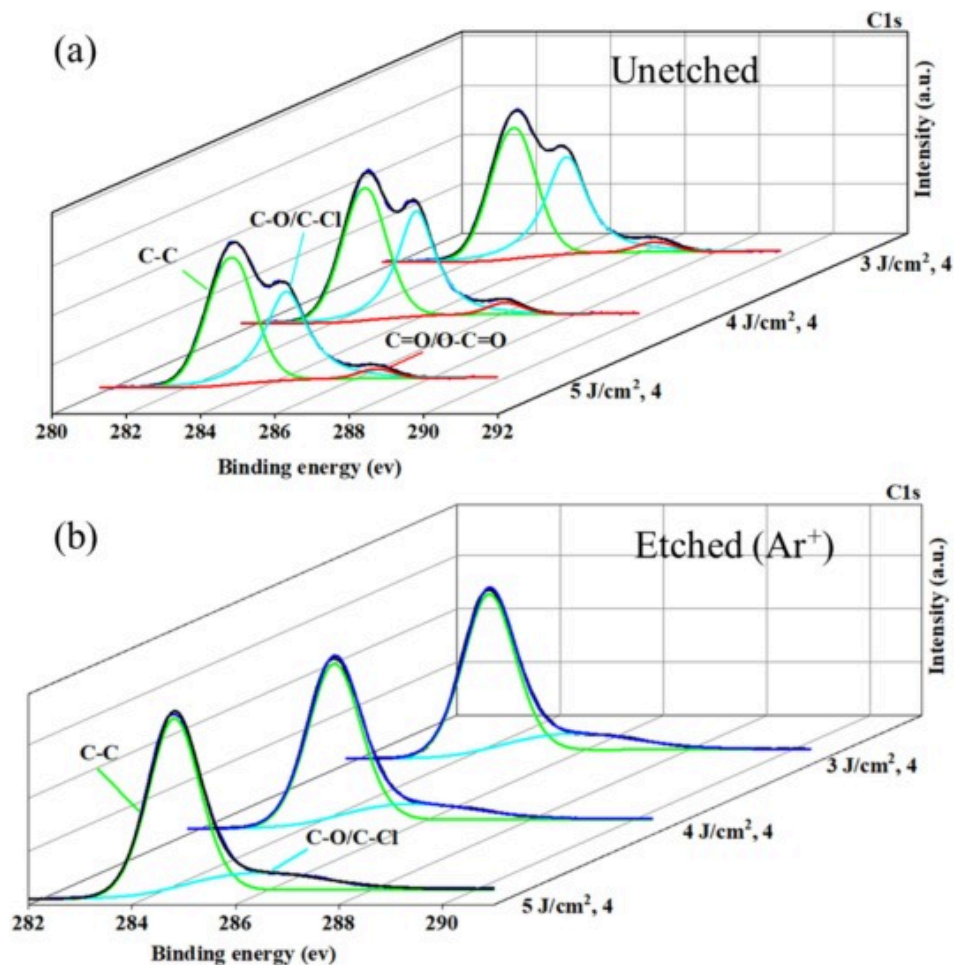
Fig. 9. Hansen solubility vector diagram of PVC/PA66 (a) without and (b) with laser ablation of PVC.

Changes in the number of generated oxygen-containing functional groups as a result of changing the laser ablation parameters can affect the miscibility of dissimilar materials. In this study, we examined the oxygen-containing functional group contents under different laser ablation parameters; the results are shown in Table 3. When $B=1$ or 4, the O/C atomic ratio increased gradually with increasing A from 1 to 7 J/cm^2 . This is because with increasing laser energy, polymers could nonlinearly absorb more photons, therefore more main chains and side chains were broken. Furthermore, thermal decomposition and photochemical reactions were intensified; therefore, more oxygen-containing functional groups were generated on the material surface. This results in an increased O/C atomic ratio. However, multiple-pulse irradiation and high-energy laser ablation could lead to slight carbonization of the material surface [37], which results in a slight decrease in the O/C atomic ratio when $A=7\text{ J/cm}^2$ and $B=4$. The thickness of the modified layer, related to the oxygen-containing functional group content, also affected the miscibility of dissimilar materials. In this study, the thickness of the modified layer was roughly estimated as follows. A 10nm thick layer of the PVC surface was removed under vacuum by Ar-ion etching, and the chemical composition of the etched surface was determined by XPS. The absorption peak of the carbon–oxygen double bond was not been

detected on the PVC surface after argon ion etching (Fig. 10), and the absorption peak of the carbon–oxygen single bond was almost disappeared. This indicates that the thickness of the laser-ablation-induced modified layer was <10nm. In summary, GFL ablation induced deposition of an oxygen-containing functional group layer of thickness <10nm on the PVC surface, and the functional group content increased with increasing laser energy. The miscibility and welding performance of dissimilar materials therefore improved.

Table 3. O/C atomic ratio of PVC.

Pulse number	O/C Atomic ratio				
	Pristine	1J/cm ²	3J/cm ²	5J/cm ²	7J/cm ²
1	(3.31%)	4.88%	6.04%	8.01%	9.87%
4	(3.31%)	6.68%	9.07%	12.60%	11.19%



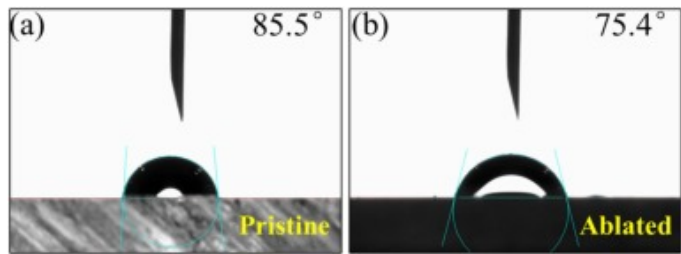
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Fig. 10. XPS C1s spectra (a) before and (b) after using argon ion etching for 180s.

Oxygen-containing functional groups are typical polar functional groups, especially –OH, which has strong hydrophilicity [38]. The WCA on the PVC surface after laser ablation decreased from 85.5° (pristine) to 75.4° (Fig. 11), that is, the hydrophilicity was enhanced, which indirectly proved that the content of oxygen-containing functional groups on the PVC surface was relatively increased. More importantly, the enhancement of hydrophilicity could be played a

positive role in improving the miscibility and welding performance between dissimilar materials.



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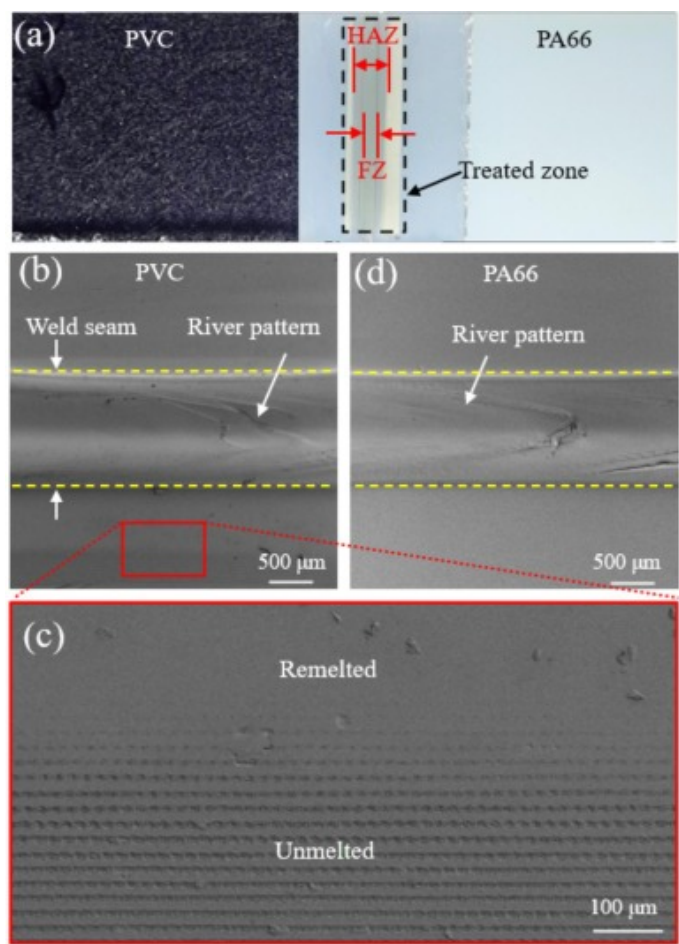
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Fig. 11. WCA on the surface of PVC samples before and after laser ablation ($4\text{J}/\text{cm}^2$, 4).

3.1.3. Joint fracture analysis

Fig. 12(a) shows that direct transmission welding of modified PVC with PA66 using a 1940nm laser without addition of a light absorber gave hybrid joints that are clean and uncontaminated. Fig. 12(b) and (c) show SEM images of the fractured PVC and PA66 surfaces, respectively, after shear tests. The images show that the pit structures generated by ablation were well preserved in areas far from the weld, but the pit structures near the weld were melted and blurred as a result of thermal effects during welding (Fig. 12(b)). This indicates that the pit structures generated by ablation disappeared during welding, failing to form the expected mechanical interlocking structure at the welding interface with PA66. Liu et al. suggested that although the surface microstructures of PC generated by ablation exhibit strong hydrophobicity, these microstructures melt and disappear during welding [15]. Consequently, it can be concluded that the surface microstructures produced on polymers (including PVC) by laser ablation might not contribute to the bonding strength of dissimilar polymers. The typical river pattern microstructures produced by ductile fracture appeared in the weld area of the fracture interface (Fig. 12(b) and (c)). This indicates tight bonding between the dissimilar materials. The SEM images prove that the microstructure generated by laser ablation did not promote the improvement of the bonding performance of dissimilar materials. However, a

combination of the SEM observation and the results described in [Section 3.1.2](#) shows that the improvement in material miscibility was the main reason for the improved weldability.



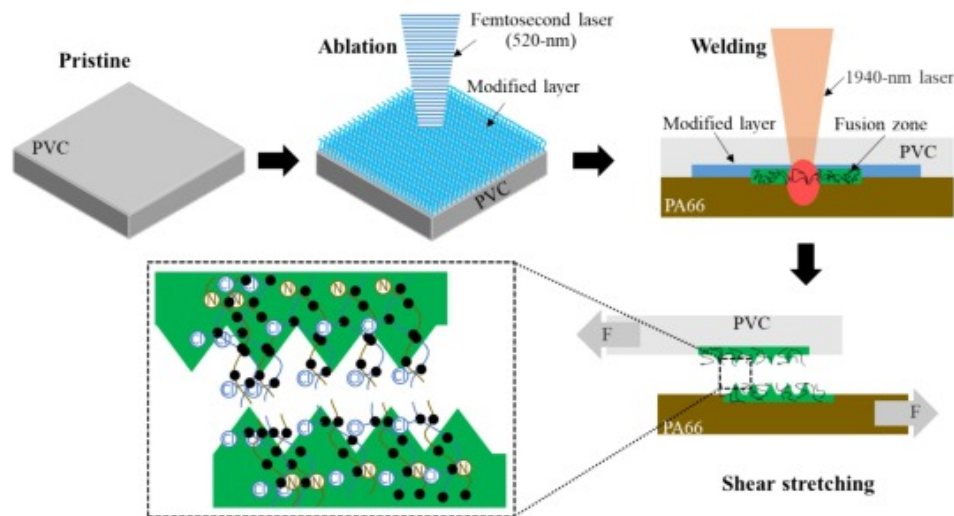
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Fig. 12. (a) Welding sample; (b) SEM image of joint fracture on PVC side; (c) SEM image of joint fracture on PA66 side.

In summary, because of synergistic photodecomposition via photothermal and photochemical effects, GFL ablation of PVC could induce generation of a large amount of carbonyl functional groups ($C=O$, $O-C=O$). This changed the polarity of the PVC surface, which improves the

miscibility and weldability of PVC/PA66, as shown in Fig. 13. In addition, the microscopic crater structures generated by laser ablation made almost no contribution to the bonding strength of hybrid joints.



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Fig. 13. Schematic diagram of joint formation and failure mechanism.

3.2. RSM experiments

3.2.1. Model data analysis

Multiple regression analysis was performed on the experimental data in Table 4; the quadratic polynomial regression equation shown in Eq. (2) was obtained. The model variance analysis results are shown in Table 5. The regression model is significant ($P < 0.01$) and the lack of fit is not significant ($P = 0.1564, > 0.05$). The adjusted model regression coefficient is 0.9364, which indicates that the model fits well and can be used to analyze and predict the bonding strength. Additionally, the significance test P -values of A , B , C , and D are all less than 0.05. This indicates that all four factors have a significant impact on the bonding strength.

(2)

$$Y = 8.10 + 0.34A + 0.52B + 0.36C - 0.53D - 0.25AB + 0.54AC - 0.14AD - 0.03BC - 0.35BD + 0.24CD - 0.29A^2 - 0.39B^2 - 1.04C^2 - 0.95D^2$$

Table 4. Experimental parameters and welding strength.

Run order	GFL Energy Density (J/cm ²)	GFL Pulse Number	Welding Power (W)	Welding Speed (mm/s)	Bonding Strength (MPa)
1	5.5	5	9	2.2	3.9
2	5.5	3	15	2.2	6.4
3	2.5	3	15	2.2	4.7
4	2.5	3	15	1.4	4.1
5	4	4	18	1.8	4.6
6	5.5	3	9	2.2	4.3
7	4	4	12	1.8	7.5
8	4	4	6	1.8	2.8
9	5.5	5	15	1.4	7.7
10	4	4	12	1	5.1
11	4	6	12	1.8	7.7
12	4	4	12	1.8	8.1
13	4	4	12	1.8	8.2
14	5.5	5	9	1.4	6.5
15	5.5	3	9	1.4	5.0
16	4	4	12	1.8	8.2
17	1	4	12	1.8	5.6
18	5.5	3	15	1.4	6.7

Run order	GFL Energy Density (J/cm ²)	GFL Pulse Number	Welding Power (W)	Welding Speed (mm/s)	Bonding Strength (MPa)
19	4	4	12	2.6	3.0
20	2.5	5	15	1.4	6.2
21	7	4	12	1.8	7.8
22	5.5	5	15	2.2	5.8
23	4	2	12	1.8	4.9
24	2.5	5	15	2.2	5.5
25	4	4	12	1.8	8.3
26	2.5	3	9	1.4	5.4
27	2.5	5	9	2.2	5.2
28	4	4	12	1.8	8.3
29	2.5	5	9	1.4	7.2
30	2.5	3	9	2.2	4.4

Table 5. Response surface analysis model evaluation results.

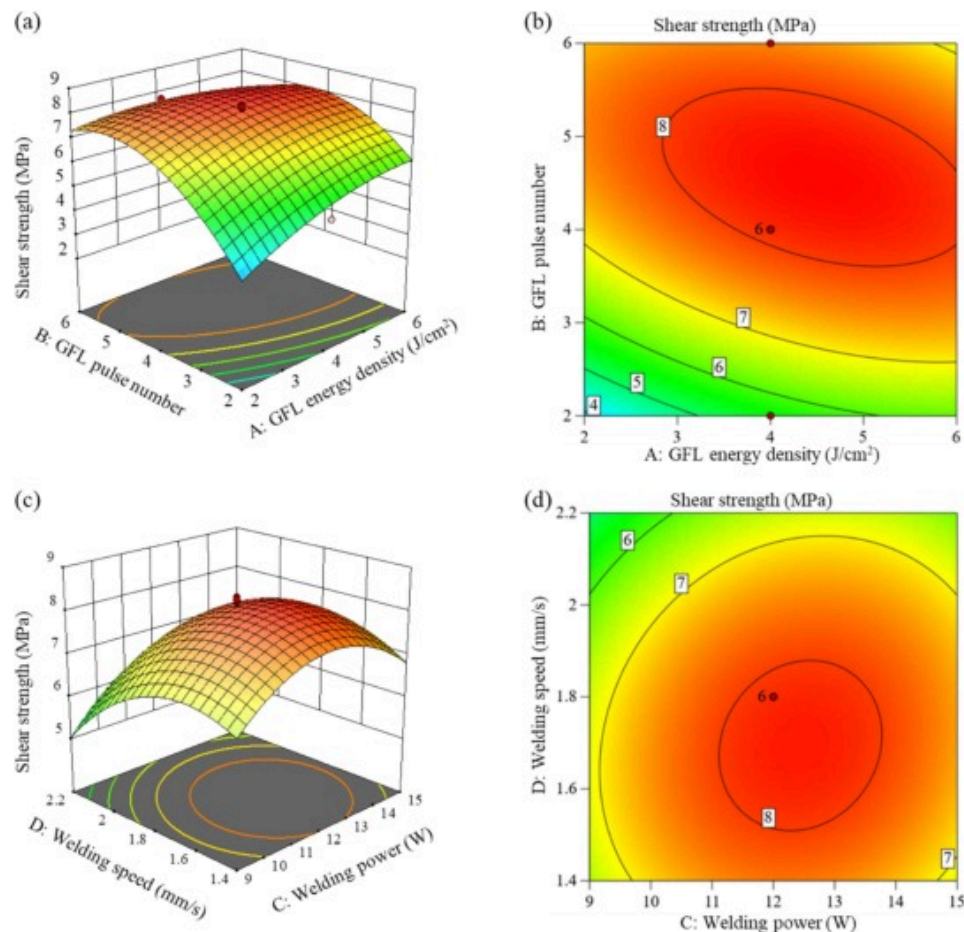
Source	Sum of Squares	df	Mean Square	F-value	P-value
Model	76.2	14	5.44	31.52	< 0.0001
A-GFL Energy Density	2.71	1	2.71	15.7	0.0013
B-GFL Pulse Number	6.48	1	6.48	37.49	< 0.0001
C-Welding Power	3.18	1	3.18	18.4	0.0006
D-Welding Speed	6.76	1	6.76	39.11	< 0.0001

Source	Sum of Squares	df	Mean Square	F-value	P-value
AB	1.03	1	1.03	5.98	0.0272
AC	4.62	1	4.62	26.76	0.0001
AD	0.3025	1	0.3025	1.75	0.2055
BC	0.0136	1	0.0136	0.0788	0.7828
BD	2.01	1	2.01	11.62	0.0039
CD	0.9025	1	0.9025	5.23	0.0372
A ²	2.26	1	2.26	13.06	0.0025
B ²	4.1	1	4.1	23.76	0.0002
C ²	29.48	1	29.48	170.71	< 0.0001
D ²	24.94	1	24.94	144.37	< 0.0001
Residual	2.59	15	0.1727		
Lack of Fit	2.17	10	0.2166	2.55	0.1564
Pure Error	0.4244	5	0.0849		
Cor Total	78.8	29	5.44	31.52	< 0.0001

3.2.2. RSM analysis

Fig. 14(a) and (b) show that the interaction between the laser energy density and pulse number significantly affected the bonding strength within the study range. The bonding strength first increased and then decreased slightly with increasing *A* and *B*, and reached a maximum at $A=4\text{--}6\text{J}/\text{cm}^2$ and $B=4\text{--}5$. In this interval, the O/C atomic ratio reached the maximum value (within the detection range), which indicates that the O/C atomic ratio has a strong correlation with the bonding strength. Specifically, the bonding strength initially increased because increasing *A* and *B* promoted the increase of O/C atomic ratio. This significantly enhanced the miscibility between dissimilar materials (see Section 3.1.2) and

therefore increased the bonding strength. However, if A and B are too large, over-ablation of the PVC surface occurred. This results in a decrease in the O/C atomic ratio and reduction of the material mechanical strength, which decreases the bonding strength of hybrid joints. The interaction between C , D , and the shear tensile strength are shown in Fig. 14(c) and (d). The maximum bonding strength was achieved with $C=11-13\text{W}$ and $D=1.6-1.8\text{mm/s}$. The welding power and scanning speed determine the laser line power density. A suitable line power density enables complete melting of the material and thorough entanglement of molecular chains, and the optimal bonding strength is achieved. According to the optimization criteria, the theoretical optimal conditions were $A=5.03\text{J/cm}^2$, $B=4.62$, $C=12.90\text{W}$, and $D=1.64\text{mm/s}$; these correspond to a bonding strength of 8.5MPa . In practice, these were corrected to $A=5.0\text{J/cm}^2$, $B=5$, $C=12.9\text{W}$, and $D=1.6\text{mm/s}$. Three parallel experiments were performed under these optimal conditions. The obtained bonding strength was 8.9MPa , which differs from the predicted value by approximately 0.4MPa .



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Fig. 14. Effects of process parameters on bonding strength: (a) 3D response surface of A, B; (b) contour map of A, B; (c) 3D response surface of C, D; (d) contour map of C, D.

4. Conclusions

In this study, we developed a surface modification method in which PVC was ablated by GFL before welding to improve the welding strength of PVC/PA66. GFL surface modification, 1940nm laser transmission welding, mechanical property testing, and related characterization

experiments were performed. The effects of the modification parameters, specifically the energy density and pulse number, on the micromorphology and chemical composition of the PVC surface were investigated. The effects of interactions among the energy density, pulse number, welding power, and welding speed on the bonding strength of hybrid joints were also studied. The failure mechanism of hybrid joints was analyzed. RSM was used to establish a mathematical model and the process parameters were optimized. The experimental results show that modification by GFL ablation effectively improved the weldability of PVC and PA66. The main conclusions are as follows.

- (1) GFL ablation changed the micromorphology of the PVC surface. Reasonable laser parameters were key to achieving effective ablation modification; the reasonable laser parameter range in this study was energy density $\geq 3 \text{ J/cm}^2$ and pulse number ≥ 3 .
- (2) Laser ablation modification induced deposition of an oxygen-containing functional group (C=O and O-C=O) layer of thickness less than 10nm on the PVC surface. The content of these functional groups generally increased with increasing laser energy. The maximum O/C ratio (12.60%) was achieved at an energy density of 5 J/cm^2 and pulse number of 4.
- (3) Increasing oxygen-containing functional group content could enhance the PVC polarity, and this improved the miscibility and weldability of PVC/PA66, and increased the bonding strength of dissimilar materials.
- (4) An RSM-based mathematical model was established. The optimal process parameter combination was determined to be energy density of 5.0 J/cm^2 , pulse number of 5, welding power of 12.9W, and welding speed of 1.6mm/s; the highest achieved bonding strength was 8.9MPa.

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CRedit authorship contribution statement

Dongyang Wang: Writing – review & editing, Writing – original draft, Investigation, Data curation, Conceptualization. **Minqiu Liu:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Sizhi Xu:** Investigation, Data curation. **Jianxun Lu:** Methodology, Funding acquisition. **Ying Zhang:** Funding acquisition, Conceptualization. **Jianpang Zhai:** Methodology, Conceptualization. **Shuangchen Ruan:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Deqin Ouyang:** Supervision, Methodology, Conceptualization. **Chunbo Li:** Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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